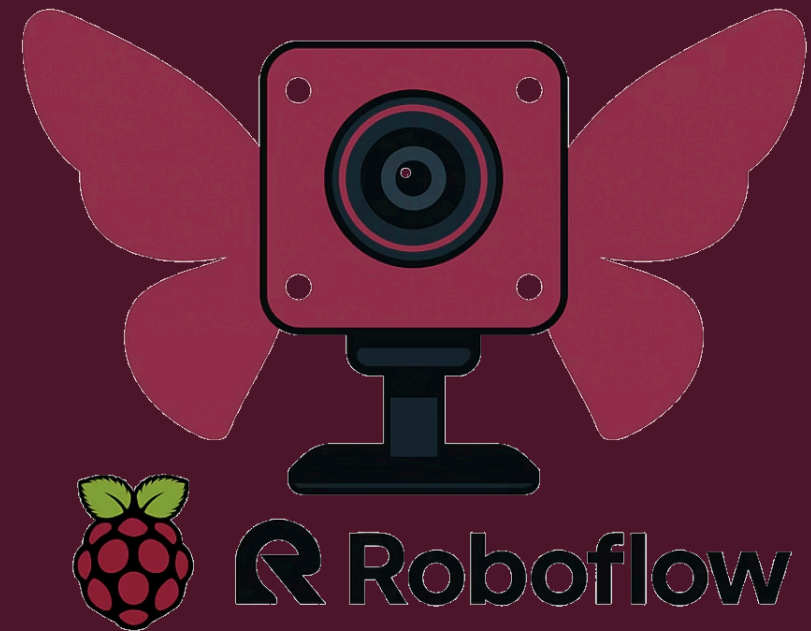


LUPI SCAN

This project detects lupus butterfly rash in facial images. It integrates data collection, model training, and deployment for real-world use.

 **by Rama ahmad**



Introduction

Lupus is an autoimmune disease that can cause visible facial symptoms, such as a butterfly-shaped rash. Early detection of these signs helps in timely diagnosis and treatment.

This project uses a **Raspberry Pi 4** to detect facial signs of lupus in real-time. A **YOLOv8 model**, trained using **Roboflow** and **Ultralytics**, performs facial detection. The system uses **OpenCV** for image processing and camera integration, providing a low-cost, portable solution for early lupus screening.

Background: Lupus and the Butterfly Rash

Key Lupus Symptom

The butterfly rash appears across cheeks and nose bridge.

Early Detection Importance

Supports timely medical screening and diagnosis stages.

Visual Challenges

Symptom varies greatly among patients, complicating detection.



Dataset Collection and Initial Size

Original Dataset

106 high-quality facial images featuring lupus signs.

Source Diversity

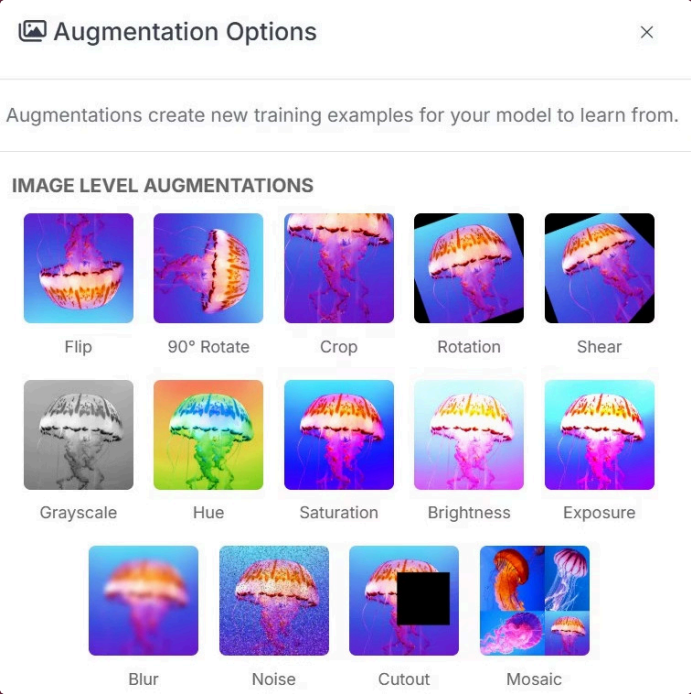
Collected from publicly available datasets from DermNet.

Data Augmentation with Roboflow

Breakdown:

178 training images
+ 51 validation images
+ 25 testing images

≤ 254 image output size



Create

Review your selections and select a version size snapshot of your dataset with the applied trans

Larger versions take longer to train but often re performance. [See how this is calculated](#)

Maximum Version Size

610 images (3x) ▼

Create

Training Split

70% of images used for training; 20% for validation and 10% for testing.

Techniques Used

- Rotation
- Scaling
- Illumination adjustment

Expanded to 610 Images

Enhanced model learning capacity with more examples.

Annotation Process

Manual Labeling

Drawn bounding boxes delineate rash-affected facial areas.

Quality Assurance

Labels confirmed clinically relevant and precise for training.



Training Process

1

We trained our project using multiple YOLO versions

YOLOv8, YOLOv11, and RF-DETR.

2

Model Evaluation Criteria

Each model was evaluated based on detection accuracy, inference speed, and training complexity.

3

YOLOv11 Performance

YOLOv11 showed incremental improvements but required more resources and longer training times.

4

RF-DETR Evaluation

RF-DETR offered excellent accuracy but was heavier and less efficient on edge devices.

5

Best Model Selection

YOLOv8 provided the best balance of speed, accuracy, and ease of deployment.

6

Final Model Decision

Based on overall performance and project requirements, we selected YOLOv8 as the final model.



Why YOLOv8 for Object Detection?



Speed & Accuracy

YOLOv8 delivers real-time object detection with high precision, ensuring both responsiveness and reliability in practical applications.



Efficient Architecture

Its single-shot detection pipeline allows for rapid inference, making it suitable for edge devices like the Raspberry Pi.



Multi-Scale Detection

The model effectively detects objects of various sizes, enabling accurate localization of skin rashes on different faces and across diverse lighting conditions.



Model Performance and Evaluation

84.2%

mAP

Mean Average Precision (mAP) is a metric used to evaluate the performance of object detection and information retrieval systems.

91.4%

Precision

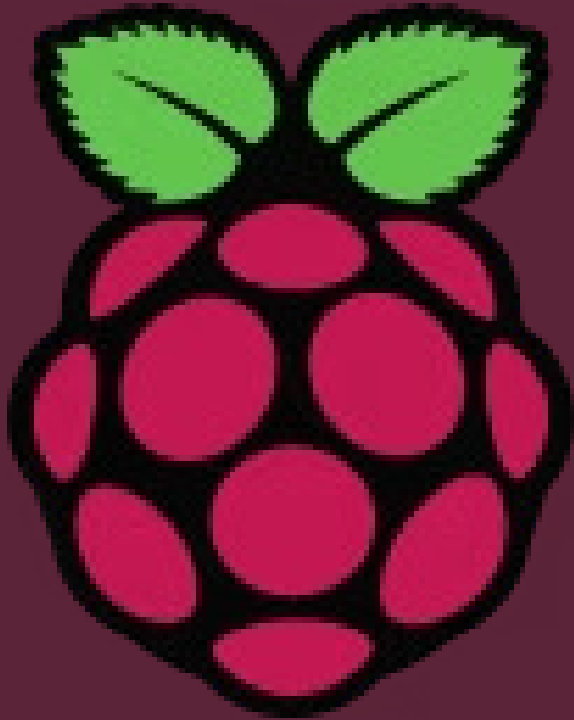
Precision indicates how many of the predicted positive cases are actually correct. High precision means few false positives.

73.2%

Recall

Recall measures how well the model can find all the actual positive instances. High recall means the model is good at minimizing false negatives.

Augmentation improved validation results, confirming stronger detection.



Embedded Deployment for Real-Time Diagnosis

- The **Raspberry Pi 4** is a compact, affordable single-board computer designed for embedded and IoT applications.
- Its small size and low power consumption make it ideal for deploying machine learning models at the edge.
- In this project, the Raspberry Pi 4 serves as the main processing unit, running the YOLOv8 model and handling real-time facial image analysis using OpenCV.

Conclusions and Limitations

1

Data Scarcity

A Limited number of labeled images of the butterfly rash, requiring augmentation.

2

Hardware Constraints

Processing limitations due to the Raspberry Pi's capabilities.

3

Environmental Variability

Accuracy may decrease under poor lighting or unconventional camera angles.

4

Model Generalization

The model is specifically trained for butterfly rash detection and may not detect other lupus manifestations.

Future Work and Improvements

1 Dataset Expansion

Increase facial image diversity (age, skin tone, disease stage) to improve model accuracy.

2 Multimodal Diagnostics

Incorporate additional inputs like symptoms, medical history, and other skin conditions.

3 Telemedicine Support

Enable real-time data sync and remote physician review.

4 Model and Hardware Optimization

Explore lightweight models and hardware accelerators for speed and efficiency.

5 Creating a Mobile App

Develop a mobile application to display detection results, store user data, and support future integration with healthcare systems.



References

Dey.S & Advisor.N. Electronic Court. Minnesota State University, Mankato A Deep Learning Model for Early Diagnosis of Systemic Lupus Erythematosus from Facial Images, 2024.



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Rama Ahmad AlJufout

Nour Hatim AlHlouI